usepackage[a4paper,margin=0.65in]geometry

[770] (CONTINUED)

ON THE 34 CONCOMITANTS OF THE TERNARY CUBIC.

Hence, the whole is

$$= (abc + 8l^3)\,\{\xi a\,(by^3 - cz^3) + \eta b\,(cz^3 - ax^3) + \zeta c\,(ax^3 - by^3)\}.$$

27. $\bar{J} = \frac{1}{4}[\text{Jac}](P, Q, A) = \frac{1}{4}\begin{vmatrix} a\xi^2 + 2l\eta\zeta, & a'\xi^2 + 2l'\eta\zeta, & x \\ b\eta^2 + 2l\zeta\xi, & b'\eta^2 + 2l'\zeta\xi, & y \\ c\zeta^2 + 2l\xi\eta, & c'\zeta^2 + 2l'\xi\eta, & z \end{vmatrix}$, if, as in a previous calculation

$$6P = a\xi^3 + b\eta^3 + c\zeta^3 + 6l\xi\eta\zeta, \qquad Q = a'\xi^3 + b'\eta^3 + c'\zeta^3 + 6l'\xi\eta\zeta.$$

Here, as before,

$$(b\eta^2+2l\zeta\xi)(c'\zeta^2+2l'\xi\eta)-(b'\eta^2+2l'\zeta\xi)(c\zeta^2+2l\xi\eta) = 2(abc+8l^3)^2\,(ca\eta^3-ab\zeta^3)\,\xi.$$

Hence, the whole is

$$= (abc + 8l^3)^2\,\{x\xi a(c\eta^3 - b\zeta^3) + y\eta b(a\xi^3 - c\zeta^3) + z\zeta c(b\zeta^3 - a\eta^3)\}.$$

20. $K = -\frac{1}{4}\{\partial_\xi\Theta\,\partial_x H + \partial_\eta\Theta\,\partial_y H + \partial_\zeta\Theta\,\partial_z H\} - 3UA$,
which, H being

$$= l^2(ax^3 + by^3 + cz^3) - (abc + 2l^3)xyz,$$

and putting

$$\Theta = (A, B, C, F, G, H)(\xi, \eta, \zeta)^2, \quad A = -l^2\xi^2 - 2al\eta\zeta, \ldots,\; F = \tfrac{1}{2}bc\xi^2 + l^2\eta\zeta,$$

$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)\,(A\xi + H\eta + G\zeta) + (by^2 + 2lzx)(H\xi + B\eta + F\zeta) + (cz^2 + 2lxy)(G\xi + F\eta + C\zeta)\}.$$

The whole coefficient of ξ is thus

$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)A + (by^2 + 2lzx)H + (cz^2 + 2lxy)G\} - (-abcl + l^4)UA$$
$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)(-l^2\xi^2 + bcyz) + (by^2 + 2lzx)(-cl^2x^2 + l^2xy) + (cz^2 + 2lxy)(-bly^2 + l^2zx)\} - (-abcl + l^4)\{ax^4 + bxy^3 + cxz^3 + 6lyzx\},$$

and herein the coefficient of x^4 is

$$= \tfrac{1}{2}a\,l^2 - al(-abc + l^3), \quad = 9al^4 - al(-abc + l^3), \quad = (abc + 8l^3)\,al,$$

viz. we have thus the term $(abc + 8l^3)\,\xi \cdot alx^4$ of the final result.

21. $K' = -(\delta)K$, where K is of the form $(abc + 8l^3)(aU + bV + cW)$; operating with (δ), we obtain $(abc + 8l^3)(a\delta U + b\delta V + c\delta W)$. Taking for instance the term of K, $(abc+8l^3)\,\xi\,[alx^4-2blxy^3-2clxz^3-3bcy^2z^2]$, then, in operating with (δ), the term bc may be considered indifferently as belonging to bV or cW, and the resulting term of K' is

$$K' = -(\delta)K = -(abc + 8l^3)\,\xi\,[al'x^4 - 2bl'xy^3 - 2cl'xz^3 + \tfrac{1}{2}\delta bc \cdot y^2z^2],$$
$$= (abc + 8l^3)\,\xi\,[(abc + 2l^3)(ax^4 - 2bxy^3 - 2cxz^3) - 18bcl^2y^2z^2].$$

28. $\bar{K} = 3\{\partial_x\Theta\,\partial_\xi P + \partial_y\Theta\,\partial_\eta P + \partial_z\Theta\,\partial_\zeta P\} + QA$; viz. writing

$\Theta = (A, B, C, F, G, H \mid \xi, \eta, \zeta)^2, \quad A = -l^2\xi^2 - 2al\eta\zeta, \ldots,\ F = \frac{1}{2}bc\xi^2 + l^2\eta\zeta, \ldots,$

then this is

$$\begin{gathered} = 3\,\{[-3lcl\xi^2 + (-abc + 4l^3)\eta\zeta]\,2\,(Ax + Hy + Gz) \\ +[-3calx^2 + (-abc + 4l^3)\zeta\xi]\,2\,(Hx + By + Fz) \\ +[-3abl\zeta^2 + (-abc + 4l^3)\xi\eta]\,2\,(Gx + Fy + Cz)\} \\ +\{(abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta\}\,(\xi x + \eta y + \zeta z). \end{gathered}$$

The whole coefficient of x is thus

$$\begin{gathered} = 3\,\{[-3lcl\xi^2 + (-abc + 4l^3)\eta\zeta]\,(Ax + Hy + Gz)\} \\ +[-3cal\eta^2 + (-abc + 4l^3)\zeta\xi]\,2\,(Hx + By + Fz) \\ +[-3abl\zeta^2 + (-abc + 4l^3)\xi\eta]\,2\,(Gx + Fy + Cz)\} \\ +\{(abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta\}\,\xi; \end{gathered}$$

herein the coefficient of ξ^4 is $18bcl^2 + (abc - 10l^3)bc$, $= (abc + 8l^3)bc$, giving, in the final result, the term $(abc + 8l^3)\,x\,bc\xi^4$.

29. $\bar{K}' = \frac{1}{2}(\delta)\bar{K}$.

Here $\bar{K}$ is of the form $(abc + 8l^3)(aU + bV + cW)$ and we have

$$\bar{K}' = \tfrac{1}{2}(abc + 8l^3)(a\delta U + b\delta V + c\delta W).$$

A term of $aU + bV + cW$ is $x\,[bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3 - 6ab\eta^2\zeta^2]$, where $bc\xi^4$ may be considered as belonging indifferently to bV or cW; and so for the other terms. The resulting term in $\frac{1}{2}(a\delta U + b\delta V + c\delta W)$ is thus

$$\tfrac{1}{2}x\,[bc'\xi^4 - 2ca'\xi\eta^3 - 2ab'\xi\zeta^3 - 6al'\eta^2\zeta^2]$$

which is

$$= x\,[l^2(bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3) + a(abc + 2l^3)\eta^2\zeta^2]$$

and we have thus a term of $\bar{K}'$.

22. $E = -\frac{1}{3}\mathrm{Jac}(\Theta, U, A)$;

Θ contains the factor $abc + 8l^3$, and if, omitting this factor, the value of Θ is called $X\xi + Y\eta + Z\zeta$, then we have

$$\begin{gathered} E = \tfrac{1}{3}\{(\xi\,\partial_x A + \eta\,\partial_x B + \zeta\,\partial_x C)(Y\zeta - Z\eta) + (\xi\,\partial_y A + \eta\,\partial_y B + \zeta\,\partial_y C)(Z\xi - X\zeta) \\ +(\xi\,\partial_z A + \eta\,\partial_z B + \zeta\,\partial_z C)(X\eta - Y\xi)\}, \end{gathered}$$

and the term herein in ξ is $-\frac{1}{3}\xi\,(Z\partial_y A - Y\partial_z A)$, where A is

$$= alx^4 - 2blxy^3 - 2clxz^3 + 3bcy^2z^2;$$

viz. the coefficient of ξ is

$$= -\tfrac{1}{3}\{(cl^2 + 2lxy)(-6blxy^2 + 6bcyz^2) - (bl^2 + 2lzx)(-6clxz^2 + 6bcy^2z)\}$$
$$= b^2cy^4z - bcl^2z^2 + 2bl^2x^2y^2 - 2cl^2x^2z^2$$
$$= (2l^2x^2 + bcyz)(by^3 - cz^3).$$

Hence, restoring the omitted factor $(abc + 8l^3)$, we have in E the term

$$(abc + 8l^3)\,\xi\,(by^3 - cz^3)\,[2l^2x^2 + bcyz].$$

23, 24. $E' = -\tfrac{1}{4}(\delta)E, \quad E'' = \tfrac{1}{8}(\delta^2)E:$

E is of the form $(abc + 8l^3)(aU + bV + cW)$, and, as before, in a term such as

$$(abc + 8l^3)\,\xi\,(by^3 - cz^3)\,(2l^2x^2 + bcyz),$$

we operate with δ or δ^2 only on the factor $2l^2x^2 + bcyz$; and in E' and E'' respectively, operating upon this factor, we obtain

$$-\tfrac{1}{4}\{4ll'x^2 + (bc' + b'c)yz\}, \quad \text{and} \quad \tfrac{1}{8}\{4l'^2x^2 + 2b'c'yz\},$$

viz. we thus obtain in E' the term

$$(abc + 8l^3)\,\xi\,(by^3 - cz^3)\,[l(abc + 2l^3)x^2 - 3bcl^2yz],$$

and in E'' the term

$$(abc + 8l^3)\,\xi\,(by^3 - cz^3)\,[(abc + 2l^3)x^2 + 18bcl^4yz].$$

30. $\bar{E} = -\tfrac{1}{3}\mathrm{Jac}(\bar{K}, U, A) \;= -\tfrac{1}{3}\begin{vmatrix} \partial_x\bar{K}, & X, & \xi \\ \partial_y\bar{K}, & Y, & \eta \\ \partial_z\bar{K}, & Z, & \zeta \end{vmatrix},$

and, if omitting in $\bar{K}$ the factor $abc + 8l^3$, we write $\bar{K} = Ax + By + Cz$, where

$$A = bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3 - 6ab\eta^2\zeta^2, \quad \text{this is} \;= -\tfrac{1}{3}\begin{vmatrix} A, & X, & \xi \\ B, & Y, & \eta \\ C, & Z, & \zeta \end{vmatrix},$$

which contains the term

$$\tfrac{1}{3}X(B\zeta - C\eta), = \tfrac{1}{3}(ax^2 + 2lyz)\,\{\zeta(ca\xi^4 - 2ab\eta\zeta^3 - 2bc\eta\xi^3 - 6bl\eta^2\zeta^2)$$
$$-\eta(abz\zeta^4 - 2bc\zeta\xi^3 - 2ca\zeta\eta^3 - 6cl\zeta^2\eta^2)\}$$
$$= (ax^2 + 2lyz)\,(c\eta^3 - b\zeta^3)\,(2l\xi^2 + a\eta\zeta).$$

Hence, restoring the factor $abc + 8l^3$, we have the terms

$$\bar{E} = (abc + 8l^3)\,\{x^2(c\eta^3 - b\zeta^3)[2al\xi^2 + a^2\eta\zeta] + yz(c\eta^3 - b\zeta^3)[4l^2\xi^2 + 2al\eta\zeta]\}.$$

31 and 32. $\bar{E}' = -\tfrac{1}{4}(\delta)\bar{E}, \quad \bar{E}'' = -\tfrac{1}{8}(\delta^2)\bar{E}:$

$\bar{E}$ is of the form $(abc + 8l^3)(aU + bV + cW)$, and we operate with δ and δ^2 on the factors $2al\xi^2 + a^2\eta\zeta$, &c.; viz.

$$\delta(2al\xi^2 + a^2\eta\zeta) = 2(a'l + al')\xi^2 + 2aa'\eta\zeta, \quad \delta^2(2al\xi^2 + a^2\eta\zeta) = 4al'^2\xi^2 + 2a'^2\eta\zeta,$$

and we thus obtain in $\bar{E}'$ the term

$$(abc + 8l^3)\, x^2\, (c\eta^3 - b\zeta^3)\, [a(abc - 4l^3)\xi^2 - 6a^2 l\eta\zeta],$$

and in $\bar{E}''$ the term

$$(abc + 8l^3)\, x^2\, (c\eta^3 - b\zeta^3)\, [-3al^2(abc + 2l^3)\xi^2 + 9a^2 l^4 \eta\zeta].$$

25. $M = \frac{1}{8}\mathrm{Jac}(U, \Psi, A)$: this, omitting the factor $(abc + 8l^3)^2$ of Ψ, is

$$= \frac{1}{8}\begin{vmatrix} ax^2 + 2lyz, & ax^2(ax^3 - 5by^3 - 5cz^3), & \xi \\ by^2 + 2lzx, & by^2(by^3 - 5cz^3 - 5ax^3), & \eta \\ cz^2 + 2lxy, & cz^2(cz^3 - 5ax^3 - 5by^3), & \zeta \end{vmatrix},$$

the coefficient of ξ herein is

$$= \tfrac{1}{8}\{(bcy^4z^2 + 2clxz^3)(cz^3 - 5ax^3 - 5by^3) - (bcy^2z^4 + 2blxy^3)(by^3 - 5cz^3 - 5ax^3)\},$$

$$= \tfrac{1}{8}\{bcy^2z^2(-by^3 + cz^3) + 2lx[-by^4 + cz^4 + 5a^2x^4(by - cz)]\},$$

$$= (by^3 - cz^3)\, [5alx^4 - blxy^3 - clxz^3 - 3bcy^2z^2].$$

Hence, restoring the factor $(abc + 8l^3)^2$, we have the term

$$(abc + 8l^3)^2\, \xi\, (by^3 - cz^3)\, [5alx^4 - blxy^3 - clxz^3 - 3bcy^2z^2].$$

26. $M' = -(\delta)M$. Here M is of the form $(abc + 8l^3)^2(a^2U + \&c.)$; and the δ operates through the $(abc + 8l^3)^2a^2$, &c.; we, in fact, have in M' the term

$$-(abc + 8l^3)^2\, \xi\, (by^3 - cz^3)\, [5al'x^4 - bl'xy^3 - cl'xz^3 - 3bc'y^2z^2],$$

which is

$$= (abc + 8l^3)^2\, \xi\, (by^3 - cz^3)\, [(abc + 2l^3)(5ax^4 - bxy^3 - cxz^3) + 18bcl^2y^2z^2].$$

33. $\bar{M} = -\frac{1}{4}[\mathrm{Jac}](P, F, A), \quad = -\frac{1}{4}\begin{vmatrix} -3lbc\xi^2 + (-abc + 4l^3)\eta\zeta, & \partial_\xi F, & x \\ -3lca\eta^2 + (-abc + 4l^3)\zeta\xi, & \partial_\eta F, & y \\ -3lab\zeta^2 + (-abc + 4l^3)\xi\eta, & \partial_\zeta F, & z \end{vmatrix},$

and the whole coefficient of x is thus

$$= \tfrac{1}{4}\{[3lca\eta^2 + (abc - 4l^3)\zeta\xi]\, \partial_\zeta F - [3lab\zeta^2 + (abc - 4l^3)\xi\eta]\, \partial_\eta F\},$$

or substituting for $\partial_\eta F$, $\partial_\zeta F$ their values, this is

$$\begin{aligned} &= \{3lca\eta^2 + (abc - 4l^3)\zeta\xi\}\, [a^2b^2\zeta^5 - (abc + 16l^3)(b\zeta^3\xi^2 + a\zeta^3\eta^2) \\ &\qquad -4l^2(bc\xi^4\eta + ca\xi^2\eta^3 + 4ab\xi\eta\zeta^2) - 8l(abc + 2l^3)\xi\eta^2\zeta] \\ &\quad -\{3lab\zeta^2 + (abc - 4l^3)\xi\eta\}\, [a^2c^2\eta^5 - (abc + 16l^3)(a\eta^3\zeta^2 + c\eta^3\xi^2) \\ &\qquad -4l^2(bc\xi^4\zeta + 4ca\xi^2\eta^2\zeta + ab\zeta^4\xi) - 8l(abc + 2l^3)\xi\eta\zeta^2]. \end{aligned}$$

Collecting, first, the terms independent of $abc - 4l^3$, and, next, those which contain $abc - 4l^3$, each set contains the factor $c\eta^3 - b\zeta^3$, and the whole is $= c\eta^3 - b\zeta^3$ multiplied by

$$-3la^3bc\eta^2\zeta^2 - 3a^2l(abc+8l^3)\eta^2\zeta^2 - 12l^3(abc\xi^4 + a^2c\eta^4 + a^2b\zeta^4) - 24a^2l^2(abc+2l^3)\xi\eta\zeta$$

$$+(abc - 4l^3)\{a^2c\xi\eta^3 + a^2b\xi\zeta^3 - (abc + 16l^3)\xi^4 + 12a^2l\eta\zeta\,\xi\}$$

and here collecting the terms in ξ^4, $\xi(c\eta^3 + b\zeta^3)$, $\xi^2\eta\zeta$ and $\eta^2\zeta^2$, each of these contains the factor $abc + 8l^3$, and, finally, the term of $\bar{M}$ is

$$= (abc+8l^3)\,(c\eta^3 - b\zeta^3)\,[(abc-8l^3)\xi^4 - a^2c\xi\eta^3 - a^2b\xi\zeta^3 - 12al^3\xi^2\eta\zeta - 6a^2bl\eta^2\zeta].$$

34. $\bar{M}' = \frac{1}{8}(\delta)\bar{M}$.

Here $\bar{M}$ is of the form $(abc + 8l^3)(aU' + bV + cW)$; and, operating with δ through the $(abc + 8l^3)\,a$, &c., we obtain in $\bar{M}'$ the term

$$\tfrac{1}{8}\,(abc + 8l^3)\,x\,(c\eta^3 - b\zeta^3)\,[(a'bc + ab'c + abc' - 24l^2l')\xi^4 + \&c.],$$

where

$$a'bc + ab'c + abc' - 24l^2l' = 18abcl^2 + 24l^2(abc + 2l^3), \quad = -7l^2(7abc + 8l^3),$$

and the term thus is

$$= (abc + 8l^3)\,x\,(c\eta^3 - b\zeta^3)\,[l^2(7abc + 8l^3)\xi^4 + \ldots].$$

This concludes the series of calculations.

Cambridge, England, 17 May, 1881.

771.

SPECIMEN OF A LITERAL TABLE FOR BINARY QUANTICS, OTHERWISE A PARTITION TABLE.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 248–255.]

The Table, commencing 1; b; c, b^2; d, bc, b^3; $\ldots$, is in fact a Partition Table, viz. considering the letters $b, c, d, \ldots$ as denoting $1, 2, 3, \ldots$ respectively, it is 1°; 1; 2, 11; 3, 12, 111; $\ldots$ a table of the partitions of the numbers $0, 1, 2, 3, \ldots$, expressed however in the literal form, in order to its giving the literal terms which enter into the coefficients of any covariant of a binary quantic. The table ought to have been made and published many years ago, before the calculation of the covariants of the quintic; and the present publication of it is, in some measure, an anachronism: but I in fact felt the need of it in some calculations in regard to the sextic; and I think the table may be found useful on other occasions. I have contented myself with calculating the table up to $z = 18$, that is, so as to include in it all the partitions of 18: it would, I think, be desirable to extend it further, say to $z = 26$; or even beyond this point, but perhaps without introducing any new letters, (that is, so as to give for the higher numbers only the partitions with a largest part not exceeding 26): the question of the space which such a table would occupy will be considered presently.

As to the employment of the table, observe that, in applying it to the case of a quantic $(a, b, c, d \mid x, y)^3$, the terms containing the letters e, f, etc., posterior to the last coefficient d of the quantic are to be disregarded; and that the terms are to be rendered homogeneous by the introduction of the proper power of the first coefficient a, rejecting any term for which the exponent of a would be negative (or what is the same thing, any term of too high a degree in the coefficients b, c, d);

thus, for the cubicovariant, where the coefficients are of the degree 3, and of the weights 3, 4, 5, 6 respectively, from the portion of the table

$$\begin{array}{cccc} d & e & f & g \\ bc & bd & be & bf \\ b^3 & c^2 & cd & ce \\ & b^2c & b^2d & d^2 \\ & & b^4 & bcd \\ & & b^4 & c^3 \\ & & & \text{etc.} \end{array}$$

we at once copy out the terms

$$\begin{array}{cccc} a^2d & abd & a^2d & ad^2 \\ abc & ac^2 & b^2d & bcd \\ b^3 & b^2c & bd^2 & d^2 \end{array}$$

which compose the coefficients in question.

As regards the formation of the table, this is at once effected, and the successive terms are obtained *currente calamo*, by Arbogast's rule of the last and the last but one: observing that each term is to be regarded as containing implicitly a power of a, so that operating on any term such as b^3, the operation on the last letter gives b^2c, and that on the last but one letter gives b^2. There is little risk of error except in the accidental omission of a term; but of course any one omission would occasion the omission of all the subsequent terms derivable from the omitted term, and would so be fatal: to remove this source of error, observe that for the successive numbers $0, 1, 2, 3$, etc., the number of partitions should be

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	...
1	1	2	3	5	7	11	15	22	30	42	56	77	101	135	176	231	297	385	...

and we can thus, for each partible number successively, verify that the right number of partitions has been obtained.

But as the number of partitions becomes large, a further control is convenient, and even necessary—say we have the 176 partitions of 15, we have by the rule to derive thence the 231 partitions of 16, and it is not until the whole of this derivation is gone through, that we could by counting the number of the new terms ascertain that the right number of 231 terms has been obtained. To break up the verification, it is convenient to know that for the partitions of 16 into 1 part, 2 parts, 3 parts, 4 parts, etc., the numbers of partitions are 1, 8, 21, 34, etc., respectively: we can then as soon as the derivations giving the partitions into 1 part, 2 parts, 3 parts, etc., respectively, have been performed, verify that the right numbers 1, 8, 21, 34, etc., of terms have been obtained. The numbers are contained in the following table, each column of which is calculated from the preceding columns according to a rule which is easily obtained, and which is itself verified by the condition that the sums of the numbers in the several columns give the before mentioned series of numbers 1, 1, 2, 3, 5, 7, etc.

No. of	Partible Number																	
Parts.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2		1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
3			1	1	2	3	4	5	7	8	10	12	14	16	19	21	24	27
4				1	1	2	3	5	6	9	11	15	18	23	27	34	39	47
5					1	1	2	3	5	7	10	13	18	23	30	37	47	57
6						1	1	2	3	5	7	11	14	20	26	35	44	58
7							1	1	2	3	5	7	11	15	21	28	38	49
8								1	1	2	3	5	7	11	15	22	29	40
9									1	1	2	3	5	7	11	15	22	30
10										1	1	2	3	5	7	11	15	22
11											1	1	2	3	5	7	11	15
12												1	1	2	3	5	7	11
13													1	1	2	3	5	7
14														1	1	2	3	5
15															1	1	2	3
16																1	1	2
17																	1	1
18																		1
	1	2	3	5	7	11	15	22	30	42	56	77	101	135	176	231	297	385

The practical rule for the construction of the table thus is:—On a sheet of paper ruled in squares, and which is read as a continuous column from the bottom of one column to the top of the next column, form the terms by Arbogast's method as already explained; writing down in pencil a batch of terms, and counting them to see that the right number has been obtained, then, at the same time verifying the derivations, mark these over in ink; and so on with another batch of terms, until the whole number of the partitions of any particular number is obtained.

The foregoing series $1, 1, 2, 3, \ldots, 385$, for the number of the partitions of the successive numbers $0, 1, 2, 3, \ldots, 18$ is carried by Euler up to the number of partitions of 59, $= 831820$, see the paper "De Partitione Numerorum," *Op. Arith. Coll.* I., bottom line of the table pp. 97–101: the continuation from the number 385 and for the partible numbers 19 to 30 is as follows:

19	20	21	22	23	24	25	26	27	28	29	30
490	627	792	1002	1255	1575	1958	2436	3010	3718	4565	5604.

the whole number of terms $1, 1, \ldots, 5604$ amounts to 28629, which at the rate of 500 to a page would occupy somewhat under 60 pages; or, at the rate here employed of 369 to a page, somewhat under 78 pages.

The Partition Table, 0 *to* 18.

Remark. The Partition Table proper of the original occupies pp. 360–364 of the volume (PDF pp. 384–388) and consists of the literal monomials representing every partition of the integers 0 through 18, arranged in columns headed by the partible number. The total tally is $1+1+2+3+5+7+11+15+22+30+42+56+77+101+135+176+231+297+385 = 1606$ distinct monomials. Owing to the extreme density of these columns (some 369 monomial entries per page), the entries are reproduced here column-by-column following Cayley's own header sequence

0; 1; 2; 3; 4; 5; 6; 7; 8; 9; 10; 11; 12; 13; 14; 15; 16; 17; 18,

each block being headed by its partible number and giving the totals

1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77, 101, 135, 176, 231, 297, 385

of monomial partitions just verified.

0: 1.

1: b.

2: c, b^2.

3: d, bc, b^3.

4: e, bd, c^2, b^2c, b^4.

5: f, be, cd, b^2d, bc^2, b^3c, b^5.

6: g, bf, ce, d^2, b^2e, bcd, c^3, b^3d, b^2c^2, b^4c, b^6.

7: h, bg, cf, de, b^2f, bce, bd^2, c^2d, b^3e, b^2cd, bc^3, b^4d, b^3c^2, b^5c, b^7.

8: i, bh, cg, df, e^2, b^2g, bcf, bde, c^2e, cd^2, b^3f, b^2ce, b^2d^2, bc^2d, c^4, b^4e, b^3cd, b^2c^3, b^5d, b^4c^2, b^6c, b^8.

9: j, bi, ch, dg, ef, b^2h, bcg, bdf, be^2, c^2f, cde, d^3, b^3g, b^2cf, b^2de, bc^2e, bcd^2, c^3d, b^4f, b^3ce, b^3d^2, b^2c^2d, bc^4, b^5e, b^4cd, b^3c^3, b^6d, b^5c^2, b^7c, b^9.

10: k, bj, ci, dh, eg, f^2, b^2i, bch, bdg, bef, c^2g, cdf, ce^2, d^2e, b^3h, b^2cg, b^2df, b^2e^2, bc^2f, $bcde$, bd^3, c^3e, c^2d^2, b^4g, b^3cf, b^3de, b^2c^2e, b^2cd^2, bc^3d, c^5, b^5f, b^4ce, b^4d^2, b^3c^2d, b^2c^4, b^6e, b^5cd, b^4c^3, b^7d, b^6c^2, b^8c, b^{10}.

(The literal tables for the partible numbers $11, 12, 13, 14, 15, 16, 17, 18$ *are reproduced in compact list form below; for the larger blocks only the leading entries are written out, the remainder being formed by Arbogast's rule from the tail of the preceding partition block, as Cayley describes above.)*

11: (56 partitions):
l, bk, cj, di, eh, fg, b^2j, bci, bdh, beg, bf^2, c^2h, cdg, cef, d^2f, de^2, b^3i, b^2ch, b^2dg, b^2ef, bc^2g, $bcdf$, bce^2, bd^2e, c^3f, c^2de, cd^3, b^4h, b^3cg, b^3df, b^3e^2, b^2c^2f, b^2cde, b^2d^3, bc^3e, bc^2d^2, c^4d, b^5g, b^4cf, b^4de, b^3c^2e, b^3cd^2, b^2c^3d, bc^5, b^6f, b^5ce, b^5d^2, b^4c^2d, b^3c^4, b^7e, b^6cd, b^5c^3, b^8d, b^7c^2, b^9c, b^{11}.

12: (77 partitions):
m, bl, ck, dj, ei, fh, g^2, b^2k, bcj, bdi, beh, bfg, c^2i, cdh, ceg, cf^2, d^2g, def, e^3, b^3j, b^2ci, b^2dh, b^2eg, b^2f^2, bc^2h, $bcdg$, $bcef$, bd^2f, bde^2, c^3g, c^2df, c^2e^2, cd^2e, d^4, b^4i, b^3ch, b^3dg, b^3ef, b^2c^2g, b^2cdf, b^2ce^2, b^2d^2e, bc^3f, bc^2de, bcd^3, c^4e, c^3d^2, b^5h, b^4cg, b^4df, b^4e^2, b^3c^2f, b^3cde, b^3d^3, b^2c^3e, $b^2c^2d^2$, bc^4d, c^6, b^6g, b^5cf, b^5de, b^4c^2e, b^4cd^2, b^3c^3d, b^2c^5, b^7f, b^6ce, b^6d^2, b^5c^2d, b^4c^4, b^8e, b^7cd, b^6c^3, b^9d, b^8c^2, $b^{10}c$, b^{12}.

13: (101 partitions):
n, bm, cl, dk, ej, fi, gh, b^2l, bck, bdj, bei, bfh, bg^2, c^2j, cdi, ceh, cfg, d^2h, deg, df^2, e^2f, b^3k, b^2cj, b^2di, b^2eh, b^2fg, bc^2i, $bcdh$, $bceg$, bcf^2, bd^2g, $bdef$, be^3, c^3h, c^2dg, c^2ef, cd^2f, cde^2, d^3e, b^4j, b^3ci, b^3dh, b^3eg, b^3f^2, b^2c^2h, b^2cdg, b^2cef, b^2d^2f, b^2de^2, bc^3g, bc^2df, bc^2e^2, bcd^2e, bd^4, c^4f, c^3de, c^2d^3, b^5i, b^4ch, b^4dg, b^4ef, b^3c^2g, b^3cdf, b^3ce^2, b^3d^2e, b^2c^3f, b^2c^2de, b^2cd^3, bc^4e, bc^3d^2, c^5d, b^6h, b^5cg, b^5df, b^5e^2, b^4c^2f, b^4cde, b^4d^3, b^3c^3e, $b^3c^2d^2$, b^2c^4d, bc^6, b^7g, b^6cf, b^6de, b^5c^2e, b^5cd^2, b^4c^3d, b^3c^5, b^8f, b^7ce, b^7d^2, b^6c^2d, b^5c^4, b^9e, b^8cd, b^7c^3, $b^{10}d$, b^9c^2, $b^{11}c$, b^{13}.

14: (135 partitions): the leading entries are
o, bn, cm, dl, ek, fj, gi, h^2, b^2m, bcl, bdk, bej, bfi, bgh, c^2k, cdj, cei, cfh, cg^2, d^2i, deh, dfg, e^2g, ef^2, b^3l, b^2ck, b^2dj, b^2ei, b^2fh, b^2g^2, bc^2j, $bcdi$, $bceh$, $bcfg$, bd^2h, $bdeg$, bdf^2, be^2f, c^3i, c^2dh, c^2eg, c^2f^2, cd^2g, $cdef$, ce^3, d^3f, d^2e^2, b^4k, b^3cj, b^3di, b^3eh, b^3fg, b^2c^2i, b^2cdh, b^2ceg, b^2cf^2, b^2d^2g, b^2def, b^2e^3, bc^3h, bc^2dg, bc^2ef, bcd^2f, $bcde^2$, bd^3e, c^4g, c^3df, c^3e^2, c^2d^2e, cd^4, etc., closing with b^{14}.

15: (176 partitions): the leading entries are
p, bo, cn, dm, el, fk, gj, hi, b^2n, bcm, bdl, bek, bfj, bgi, bh^2, c^2l, cdk, cej, cfi, cgh, d^2j, dei, dfh, dg^2, e^2h, efg, f^3, b^3m, b^2cl, b^2dk, b^2ej, b^2fi, b^2gh, bc^2k, $bcdj$, $bcei$, $bcfh$, bcg^2, bd^2i, $bdeh$, $bdfg$, be^2g, bef^2, c^3j, c^2di, c^2eh, c^2fg, cd^2h, $cdeg$, cdf^2, ce^2f, d^3g, d^2ef, de^3, etc., closing with b^{15}.

16: (231 partitions): the leading entries are
q, bp, co, dn, em, fl, gk, hj, i^2, b^2o, bcn, bdm, bel, bfk, bgj, bhi, c^2m, cdl, cek, cfj, cgi, ch^2, d^2k, dej, dfi, dgh, e^2i, efh, eg^2, f^2g, b^3n, b^2cm, b^2dl, b^2ek, b^2fj, b^2gi, b^2h^2, bc^2l, $bcdk$, $bcej$, $bcfi$, $bcgh$, bd^2j, $bdei$, $bdfh$, bdg^2, be^2h, $befg$, bf^3, c^3k, c^2dj, c^2ei, c^2fh, c^2g^2, cd^2i, $cdeh$, $cdfg$, ce^2g, cef^2, d^3h, d^2eg, d^2f^2, de^2f, e^4, etc., closing with b^{16}.

17: (297 partitions): the leading entries are
r, bq, cp, do, en, fm, gl, hk, ij, b^2p, bco, bdn, bem, bfl, bgk, bhj, bi^2, c^2n, cdm, cel, cfk, cgj, chi, d^2l, dek, dfj, dgi, dh^2, e^2j, efi, egh, f^2h, fg^2, etc., closing with b^{17}.

18: (385 partitions): the leading entries are
s, br, cq, dp, eo, fn, gm, hl, ik, j^2, b^2q, bcp, bdo, ben, bfm, bgl, bhk, bij, c^2o, cdn, cem, cfl, cgk, chj, ci^2, d^2m, del, dfk, dgj, dhi, e^2k, efj, egi, eh^2, f^2i, fgh, g^3, etc., closing with b^{18}.

Note.—In the tables on this page, a has been treated like the other letters; on the preceding pages, the powers of a have been suppressed except in the first of every series of terms containing a common power of a. The complete column-by-column literal listings of Cayley's original (*Amer. J. Math.*, vol. iv, pp. 248–255) are voluminous; in this transcription only the leading sections of the tables for 16, 17, 18 have been written out in full—the remainder being formed by Arbogast's rule from the tail of the preceding partition block as Cayley himself describes above.

772.

ON THE ANALYTICAL FORMS CALLED TREES.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 266–268.]

In a tree of N knots, selecting any knot at pleasure as a root, the tree may be regarded as springing from this root, and it is then called a root-tree. The same tree thus presents itself in various forms as a root-tree; and if we consider the different root-trees with N knots, these are not all of them distinct trees. We have thus the two questions, to find the number of root-trees with N knots; and, to find the number of distinct trees with N knots.

I have in my paper "On the Theory of the Analytical Forms called Trees," *Phil. Mag.*, t. xiii. (1857), pp. 172–176, [203] given the solution of the first question; viz. if ϕ_N denotes the number of the root-trees with N knots, then the successive numbers ϕ_1, ϕ_2, ϕ_3, etc., are given by the formula

$$\phi_1 + x\phi_2 + x^2\phi_3 + \ldots = (1-x)^{-\phi_1}(1-x^2)^{-\phi_2}(1-x^3)^{-\phi_3}\ldots,$$

viz. we thus find

suffix of ϕ	1	2	3	4	5	6	7	8	9	10	11	12	13
$\phi =$	1	1	2	4	9	20	48	115	286	719	1842	4766	12486.

And I have, in the paper "On the analytical forms called Trees, with application to the theory of chemical combinations," *Brit. Assoc. Report*, 1875, pp. 257–305, [610] also shown how by the consideration of the centre or bicentre "of length" we can obtain formulae for the number of central and bicentral trees, that is, for the number of distinct trees, with N knots: the numerical result obtained for the total number of distinct trees with N knots is given as follows:

No. of Knots	1	2	3	4	5	6	7	8	9	10	11	12	13
No. of Central Trees	1	0	1	1	2	3	7	12	27	55	127	284	682
Bicentral	0	1	0	1	1	3	4	11	20	51	108	267	619
Total	1	1	1	2	3	6	11	23	47	106	235	551	1301.

But a more simple solution is obtained by the consideration of the centre or bicentre "of number." A tree of an odd number N of knots has a centre of number, and a tree of an even number N of knots has a centre or else a bicentre of number. To explain this notion (due to M. Camille Jordan) we consider the branches which proceed from any knot, and (excluding always this knot itself) we count the number of the knots upon the several branches; say these numbers are $\alpha, \beta, \gamma, \delta, \epsilon$, etc., where of course $\alpha + \beta + \gamma + \delta + \epsilon + \text{etc.} = N - 1$. If N is even we may have, say $\alpha = \frac{1}{2}N$; and then $\beta + \gamma + \delta + \epsilon + \text{etc.} = \frac{1}{2}N - 1$, viz. α is larger by unity than the sum of the remaining numbers: the branch with α knots, or the number α, is said to be "merely dominant." If N be odd, we cannot of course have $\alpha = \frac{1}{2}N$, but we may have $\alpha > \frac{1}{2}N$; here α exceeds by 2 at least the sum of the other numbers; and the branch with α knots, or the number α, is said to be "predominant." In every other case, viz. in the case where each number α is less than $\frac{1}{2}N$, (and where consequently the largest number α does not exceed the sum of the remaining numbers), the several branches, or the numbers α, β, γ, etc., are said to be subequal. And we have the theorem. First, when N is odd, there is always one knot (and only one knot) for which the branches are subequal: such knot is called the centre of number. Secondly, when N is even, either there is one knot (and only one knot) for which the branches are subequal: and such knot is then called the centre of number; or else there is no such knot, but there are two adjacent knots (and no other knot) each having a merely-dominant branch: such two knots are called the bicentre of number, and each of them separately is a half-centre.

Considering now the trees with N knots as springing from a centre or a bicentre of number, and writing ψ_N for the whole number of distinct trees with N knots, we readily obtain these in terms of the foregoing numbers ϕ_1, ϕ_2, ϕ_3, etc., viz. we have

$$\begin{aligned}
\psi_1 &= 1, \\
\psi_2 &= \text{coeff. } x^0 \text{ in } (1-x)^{-\phi_1}, \\
\psi_3 &= \text{coeff. } x^2 \text{ in } (1-x)^{-\phi_1}, \\
\psi_4 &= \tfrac{1}{2}\phi_2(\phi_2+1) + \text{coeff. } x^3 \text{ in } (1-x)^{-\phi_1}, \\
\psi_5 &= \text{coeff. } x^4 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2}, \\
\psi_6 &= \tfrac{1}{2}\phi_3(\phi_3+1) + \text{coeff. } x^5 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2},
\end{aligned}$$

$$\psi_7 = \text{coeff. } x^6 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2}(1-x^3)^{-\phi_3},$$

and so on, the law being obvious. And the formulae are at once seen to be true. Thus for $N = 6$, the formula is

$$\psi_6 = \tfrac{1}{2}\phi_3(\phi_3+1) + \tfrac{1}{2}\phi_2(\phi_2+1)\cdot\phi_1 + \phi_2\cdot\tfrac{1}{2}\phi_1(\phi_1+1)(\phi_1+2)$$
$$+\tfrac{1}{120}\phi_1(\phi_1+1)(\phi_1+2)(\phi_1+3)(\phi_1+4).$$

We have ϕ_3 root-trees with 3 knots, and by simply joining together any two of them, treating the two roots as a bicentre, we have all the bicentral trees with 6 knots: this accounts for the term $\frac{1}{2}\phi_3(\phi_3+1)$. Again, we have ϕ_1 root-trees with 1 knot, ϕ_2 root-trees with 2 knots; and with a given knot as centre, and the partitions $(2,2,1)$, $(2,1,1,1)$, $(1,1,1,1,1)$ successively, we build up the central trees of 6 knots, viz. 1° we take as branches any two ϕ_2's and any one ϕ_1; 2° any one ϕ_2 and any three ϕ_1's; 3° any five ϕ_1's; the partitions in question being all the partitions of 5 with no part greater than 2, that is, all the partitions with subequal parts. We easily obtain

suffix of ψ	1	2	3	4	5	6	7	8	9	10	11	12	13
$\psi =$	1	1	1	2	3	6	11	23	47	106	235	551	1301

agreeing with the results obtained by the much more complicated formulae of the paper of 1875.

773.

ON THE 8-SQUARE IMAGINARIES.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 293–296.]

I write throughout 0 to denote positive unity, and uniting with it the seven imaginaries $1, \dots, 7$, form an octavic system $0, 1, 2, 3, 4, 5, 6, 7$, the laws of combination being

$$0^2 = 0, \qquad 1^2 = 2^2 = 3^2 = 4^2 = 5^2 = 6^2 = 7^2 = -0,$$

$$123 = \epsilon_1, \qquad 145 = \epsilon_2, \qquad 167 = \epsilon_3,$$

$$246 = \epsilon_4, \qquad 257 = \epsilon_5,$$

$$347 = \epsilon_6, \qquad 356 = \epsilon_7,$$

where $\epsilon = \pm$, viz. each ϵ has a determinate value $+$ or $-$ as the case may be; and where the formula $123 = \epsilon_1$ denotes the six equations

$$23 = \epsilon_1 \cdot 1, \qquad 31 = \epsilon_1 \cdot 2, \qquad 12 = \epsilon_1 \cdot 3,$$

$$32 = -\epsilon_1 \cdot 1, \quad 13 = -\epsilon_1 \cdot 2, \quad 21 = -\epsilon_1 \cdot 3,$$

and so for the other formulae.

The multiplication table of the eight symbols thus is

	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	-0	$\epsilon_1 3$	$-\epsilon_1 2$	$\epsilon_2 5$	$-\epsilon_2 4$	$\epsilon_3 7$	$-\epsilon_3 6$
2	2	$-\epsilon_1 3$	-0	$\epsilon_1 1$	$\epsilon_4 6$	$\epsilon_5 7$	$-\epsilon_4 4$	$-\epsilon_5 5$
3	3	$\epsilon_1 2$	$-\epsilon_1 1$	-0	$\epsilon_6 7$	$\epsilon_7 6$	$-\epsilon_7 5$	$-\epsilon_6 4$
4	4	$-\epsilon_2 5$	$-\epsilon_4 6$	$-\epsilon_6 7$	-0	$\epsilon_2 1$	$\epsilon_4 2$	$\epsilon_6 3$
5	5	$\epsilon_2 4$	$-\epsilon_5 7$	$-\epsilon_7 6$	$-\epsilon_2 1$	-0	$\epsilon_7 3$	$\epsilon_5 2$
6	6	$-\epsilon_3 7$	$\epsilon_4 4$	$\epsilon_7 5$	$-\epsilon_4 2$	$-\epsilon_7 3$	-0	$\epsilon_3 1$
7	7	$\epsilon_3 6$	$\epsilon_5 5$	$\epsilon_6 4$	$-\epsilon_6 3$	$-\epsilon_5 2$	$-\epsilon_3 1$	-0

Hence if $0, 1, 2, 3, 4, 5, 6, 7$ and $0', 1', 2', 3', 4', 5', 6', 7'$ denote ordinary algebraical magnitudes, and we form the product

$$(00+11+22+33+44+55+66+77)\,(0'0+1'1+2'2+3'3+4'4+5'5+6'6+7'7),$$

this is at once found to be $=$

$$\begin{aligned}
&(00' - 11' - 22' - 33' - 44' - 55' - 66' - 77')\,0\\
&\quad +(01' + 0'1 + \epsilon_1\overline{23} + \epsilon_2\overline{45} + \epsilon_3\overline{67})\,1\\
&\quad +(02' + 0'2 + \epsilon_1\overline{31} + \epsilon_4\overline{46} + \epsilon_5\overline{57})\,2\\
&\quad +(03' + 0'3 + \epsilon_1\overline{12} + \epsilon_6\overline{47} + \epsilon_7\overline{56})\,3\\
&\quad +(04' + 0'4 + \epsilon_2\overline{51} + \epsilon_4\overline{62} + \epsilon_6\overline{73})\,4\\
&\quad +(05' + 0'5 + \epsilon_2\overline{14} + \epsilon_5\overline{72} + \epsilon_7\overline{63})\,5\\
&\quad +(06' + 0'6 + \epsilon_3\overline{71} + \epsilon_4\overline{24} + \epsilon_7\overline{35})\,6\\
&\quad +(07' + 0'7 + \epsilon_3\overline{16} + \epsilon_5\overline{25} + \epsilon_6\overline{34})\,7,
\end{aligned}$$

where $\overline{12}$ is written to denote $12' - 1'2$, and so in other cases.

The sum of the squares of the eight coefficients of $0, 1, 2, 3, 4, 5, 6, 7$ respectively will, if certain terms destroy each other, be

$$= (0^2+1^2+2^2+3^2+4^2+5^2+6^2+7^2)(0'^2+1'^2+2'^2+3'^2+4'^2+5'^2+6'^2+7'^2)$$

viz. the sum of the squares contains the several terms

$$\begin{array}{lllllll}
\epsilon_1\epsilon_2\overline{23}.\overline{45}, & \epsilon_1\epsilon_3\overline{23}.\overline{67}, & \epsilon_1\epsilon_4\overline{31}.\overline{46}, & \epsilon_1\epsilon_5\overline{31}.\overline{57}, & \epsilon_1\epsilon_6\overline{12}.\overline{47}, & \epsilon_1\epsilon_7\overline{12}.\overline{56}, & \epsilon_2\epsilon_3\overline{45}.\overline{67},\\
\epsilon_4\epsilon_5\overline{24}.\overline{35}, & \epsilon_4\epsilon_6\overline{62}.\overline{73}, & \epsilon_2\epsilon_4\overline{14}.\overline{63}, & \epsilon_2\epsilon_7\overline{51}.\overline{73}, & \epsilon_2\epsilon_5\overline{14}.\overline{72}, & \epsilon_2\epsilon_6\overline{51}.\overline{62}, & \epsilon_4\epsilon_7\overline{46}.\overline{57},\\
\epsilon_3\epsilon_4\overline{25}.\overline{34}, & \epsilon_3\epsilon_7\overline{72}.\overline{63}, & \epsilon_6\epsilon_7\overline{16}.\overline{34}, & \epsilon_3\epsilon_5\overline{71}.\overline{35}, & \epsilon_3\epsilon_6\overline{71}.\overline{24}, & \epsilon_5\epsilon_6\overline{16}.\overline{25}, & \epsilon_5\epsilon_7\overline{47}.\overline{56},
\end{array}$$

and observing that $\overline{21} = -\overline{12}$, etc., and that we have identically

$$\overline{23}.\overline{45} + \overline{24}.\overline{53} + \overline{25}.\overline{34} = \text{zero}, \quad \text{etc.},$$

then the three terms of each column will vanish, provided a proper relation exists between the ϵ's: viz. the conditions which we thus obtain are

$$\begin{aligned}
\epsilon_1\epsilon_2 &= \epsilon_4\epsilon_6 = \epsilon_5\epsilon_7,\\
\epsilon_1\epsilon_3 &= \epsilon_3\epsilon_6 = \epsilon_2\epsilon_7,\\
\epsilon_1\epsilon_5 &= \epsilon_3\epsilon_7 = \epsilon_2\epsilon_6,\\
\epsilon_1\epsilon_6 &= \epsilon_3\epsilon_5 = \epsilon_3\epsilon_4,\\
\epsilon_1\epsilon_7 &= -\epsilon_2\epsilon_4 = \epsilon_3\epsilon_5,
\end{aligned}$$

$$\epsilon_2\epsilon_3 = -\epsilon_4\epsilon_6 = \epsilon_4\epsilon_7.$$

We may without loss of generality assume $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$; the equations then become

$$+ = -\epsilon_4\epsilon_7 = \epsilon_5\epsilon_6,$$

$$+ = -\epsilon_4\epsilon_6 = \epsilon_5\epsilon_7,$$

$$+ = -\epsilon_4\epsilon_5 = \epsilon_6\epsilon_7,$$

$$\epsilon_4 = \epsilon_5 = \epsilon_6,$$

$$\epsilon_5 = \epsilon_7 = \epsilon_6,$$

$$\epsilon_6 = \epsilon_5 = -\epsilon_4,$$

$$\epsilon_7 = -\epsilon_4 = \epsilon_5,$$

and writing $\theta = \pm$ at pleasure, these are all satisfied if $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$. The terms written down all disappear, and the sum of the squares of the eight coefficients thus becomes equal to the product of two sums each of them of eight squares, viz. this is the case if $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$, $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$, θ being $= \pm$ at pleasure: the resulting system of imaginaries may be said to be an 8-square system.

We may inquire whether the system is associative; for this purpose, supposing in the first instance that the ϵ's remain arbitrary, we form the complete system of the values of the triplets 12.3, 1.23, etc., (read the top line $12.3 = -\epsilon_1 0$, $1.23 = -\epsilon_1 0$, the next line $12.4 = \epsilon_1\epsilon_6 7$, $1.24 = \epsilon_2\epsilon_4 7$, and so in other cases):

$$
\begin{array}{lll}
12.3 = 1.23 = & -\epsilon_1 & -\epsilon_1 \cdot 0 \\
12.4 = 1.24 = & \epsilon_1\epsilon_6 & \epsilon_2\epsilon_4 \cdot 7 \\
12.5 = 1.25 = & \epsilon_1\epsilon_7 & -\epsilon_2\epsilon_5 \cdot 6 \\
12.6 = 1.26 = & -\epsilon_1\epsilon_7 & -\epsilon_4\epsilon_2 \cdot 5 \\
12.7 = 1.27 = & -\epsilon_1\epsilon_6 & \epsilon_3\epsilon_2 \cdot 4 \\
13.4 = 1.34 = & -\epsilon_1\epsilon_4 & -\epsilon_3\epsilon_6 \cdot 6 \\
13.5 = 1.35 = & -\epsilon_1\epsilon_5 & \epsilon_3\epsilon_7 \cdot 7 \\
13.6 = 1.36 = & \epsilon_1\epsilon_4 & \epsilon_3\epsilon_7 \cdot 4 \\
13.7 = 1.37 = & \epsilon_1\epsilon_5 & -\epsilon_3\epsilon_6 \cdot 5 \\
14.5 = 1.45 = & -\epsilon_2 & -\epsilon_2 \cdot 0 \\
14.6 = 1.46 = & \epsilon_2\epsilon_4 & \epsilon_1\epsilon_4 \cdot 3 \\
14.7 = 1.47 = & \epsilon_2\epsilon_6 & -\epsilon_1\epsilon_6 \cdot 2 \\
15.6 = 1.56 = & -\epsilon_2\epsilon_4 & -\epsilon_1\epsilon_7 \cdot 2 \\
15.7 = 1.57 = & -\epsilon_2\epsilon_5 & \epsilon_1\epsilon_5 \cdot 3 \\
16.7 = 1.67 = & -\epsilon_3 & -\epsilon_3 \cdot 0 \\
23.4 = 2.34 = & \epsilon_1\epsilon_6 & -\epsilon_4\epsilon_6 \cdot 5 \\
23.5 = 2.35 = & -\epsilon_1\epsilon_7 & -\epsilon_4\epsilon_7 \cdot 4 \\
23.6 = 2.36 = & \epsilon_1\epsilon_4 & -\epsilon_5\epsilon_7 \cdot 7 \\
23.7 = 2.37 = & -\epsilon_1\epsilon_5 & -\epsilon_4\epsilon_5 \cdot 6 \\
24.5 = 2.45 = & -\epsilon_4\epsilon_6 & -\epsilon_5\epsilon_7 \cdot 3 \\
24.6 = 2.46 = & -\epsilon_4 & -\epsilon_4 \cdot 0 \\
24.7 = 2.47 = & \epsilon_4\epsilon_6 & \epsilon_1\epsilon_6 \cdot 1 \\
25.6 = 2.56 = & -\epsilon_4\epsilon_5 & \epsilon_1\epsilon_7 \cdot 1 \\
25.7 = 2.57 = & -\epsilon_5 & -\epsilon_5 \cdot 0 \\
26.7 = 2.67 = & -\epsilon_4\epsilon_6 & -\epsilon_1\epsilon_3 \cdot 3 \\
34.5 = 3.45 = & -\epsilon_6\epsilon_7 & \epsilon_1\epsilon_2 \cdot 2 \\
34.6 = 3.46 = & -\epsilon_6\epsilon_7 & -\epsilon_1\epsilon_4 \cdot 1 \\
34.7 = 3.47 = & -\epsilon_6 & -\epsilon_6 \cdot 0 \\
35.6 = 3.56 = & -\epsilon_7 & -\epsilon_7 \cdot 0 \\
35.7 = 3.57 = & \epsilon_5\epsilon_7 & -\epsilon_1\epsilon_5 \cdot 1 \\
36.7 = 3.67 = & -\epsilon_6\epsilon_7 & \epsilon_1\epsilon_3 \cdot 2 \\
45.6 = 4.56 = & \epsilon_2\epsilon_3 & -\epsilon_6\epsilon_7 \cdot 7 \\
45.7 = 4.57 = & -\epsilon_2\epsilon_3 & -\epsilon_4\epsilon_5 \cdot 6 \\
46.7 = 4.67 = & -\epsilon_4\epsilon_6 & -\epsilon_2\epsilon_3 \cdot 5 \\
56.7 = 5.67 = & -\epsilon_5\epsilon_7 & \epsilon_2\epsilon_6 \cdot 4.
\end{array}
$$

Write as before $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$; then, disregarding the lines (such as the first line) which contain the symbol 0, and writing down only the signs as given in the third and fourth columns, these are

$$
\begin{array}{cc}
\epsilon_6 & \epsilon_4 \\
\epsilon_7 & -\epsilon_5 \\
-\epsilon_7 & -\epsilon_4 \\
-\epsilon_6 & \epsilon_5 \\
-\epsilon_4 & -\epsilon_6 \\
-\epsilon_5 & \epsilon_7 \\
\epsilon_4 & \epsilon_7 \\
\epsilon_5 & -\epsilon_6 \\
\epsilon_7 & \epsilon_4 \\
\epsilon_5 & -\epsilon_6 \\
-\epsilon_4 & -\epsilon_7 \\
-\epsilon_5 & \epsilon_5 \\
+ & -\epsilon_6\epsilon_7 \\
 & -\epsilon_4\epsilon_5 \\
+ & -\epsilon_5\epsilon_7 \\
- & -\epsilon_4\epsilon_6 \\
-\epsilon_4\epsilon_7 & - \\
\epsilon_4 & \epsilon_6 \\
-\epsilon_5 & \epsilon_7 \\
-\epsilon_4\epsilon_6 & - \\
-\epsilon_5\epsilon_6 & + \\
-\epsilon_6 & -\epsilon_4 \\
\epsilon_7 & -\epsilon_5 \\
-\epsilon_5\epsilon_7 & + \\
+ & -\epsilon_6\epsilon_7 \\
- & -\epsilon_4\epsilon_5 \\
-\epsilon_4\epsilon_5 & - \\
-\epsilon_6\epsilon_7 & +
\end{array}
$$

We hence see at once that the pairs of signs in the two columns respectively cannot be made identical: to make them so, we should have $\epsilon_6 = \epsilon_4$, $\epsilon_7 = -\epsilon_4$, $\epsilon_7 = \epsilon_4$, that is, $\epsilon_4 = \epsilon_6 = \epsilon_7 = -\epsilon_5$, which is inconsistent with the last equation of the system $-\epsilon_6\epsilon_7 = +$.

Hence the imaginaries $1, 2, 3, 4, 5, 6, 7$, as defined by the original conditions, are not in any case associative.

If we have $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$ and also $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$, that is, if the imaginaries belong to the 8-square formula, then it is at once seen that each pair consists of two opposite signs; that is, for the several triads 123, 145, 167, 246, 257, 347, 356 used for the definition of the imaginaries, the associative property holds good, $12.3 = 1.23$, etc.; but for each of the remaining twenty-eight triads, the two terms *are equal but of opposite signs*, viz. $12.4 = -1.24$, etc.; so that the product 124 of any such three symbols has no determinate meaning.

Baltimore, March 5th, 1882.

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TABLES FOR THE BINARY SEXTIC.

The Leading Coefficients of the First 18 of the 26 Covariants.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 379–384.]

Including the sextic itself, the number of covariants of the binary sextic is = 26, as shown in the table p. 296 of Clebsch's *Theorie der binären algebraischen Formen*, Leipzig, 1872; viz. this is

	Order					
Deg.	2	4	6	8	10	12
1			f			
2	A		i		H	
3		l		p	$(i,f)_1$	T
4	B		$(i,f)_2$	$(f,l)_1$	$(p,f)_1$	$(B,i)_1$
5			$(i,l)_3$	$(i,l)_2$	$(H,l)_1$	
6	A_2		$(p,l)_2$			
7		$(l,p)_4$	$(l,l)_3$			
8		$(i,l^2)_4$				
9			$((l,l)_2,l)_4$			
10	$(l,l^2)_6$	$(l,l^2)_5$				
12		$(l\cdot i)_6$				
15	$((l,l)_2,l)_6$					

Or, using the capital letters $A, B, \ldots, Z$ to denote the 26 covariants in the same order, the table is

	2	4	6	8	10	12
1			A			
2	B		C		D	
3		E		F	$G, = (A, C)^2$	H
4	I		$J, = (A, E)^2$	$K, = (A, E)^3$	$L, = (D, C)^2$	
5		$M, = (C, E)^2$	$N, = (C, E)^3$	$O, = (D, E)^2$		
6	P		$R, = (C, E^2)^4$			
7		$S, = (A, E^2)^2$	$T, = (A, E^2)^3$			
8		$U, = (C, E^2)^4$				
9			$V, = (C, E^2)^4$			
10	$W, = (A, E^3)^6$	$X, = (A, E^3)^5$				
12		$Y, = (C, E^3)^6$				
15	$Z, = (C, E^3)^6$					

A is the sextic.
B is Salmon's A, p. 202.
I „ „ B, p. 203.
P is Salmon's C, p. 204.
W, „ „ C, p. 207.
Z, „ E, p. 253.
The references are to Salmon's *Higher Algebra*, 2nd Ed., 1866.

In the present short paper I give the leading coefficients of the first 18 covariants, A to R (some of these are of course known values, but it is convenient to include them): for the next four covariants S, T, U, V, the leading coefficients depend upon the coefficients of A, C, G and E^2, viz. writing

$$A = (a, b, c, d, e, f, g \mid x, y)^6,$$
$$E^2 = (0, \tfrac{1}{2}\beta, \tfrac{1}{4}\gamma, \tfrac{1}{6}\delta, \tfrac{1}{4}\epsilon, \tfrac{1}{2}\zeta \mid x, y)^6,$$
$$C = (\alpha', \tfrac{1}{2}\beta', \tfrac{1}{4}\gamma', \tfrac{1}{6}\delta', \tfrac{1}{4}\epsilon', \ldots \mid x, y)^6,$$
$$G = (\alpha'', \tfrac{1}{2}\beta'', \tfrac{1}{4}\gamma'', \tfrac{1}{6}\delta'', \tfrac{1}{4}\epsilon'', \ldots \mid x, y)^6,$$

we have

$$S, \text{ Coeff. } x^4 = a\epsilon - b\delta + c\gamma - d\beta + e\alpha,$$
$$T, \text{ „ „ } x^6 = a\delta - 2b\gamma + 3c\beta - 4d\alpha,$$
$$U, \text{ „ „ } x^4 = 2\alpha'\delta - \beta'\gamma + \delta'\beta - 2\epsilon'\alpha,$$
$$V, \text{ „ „ } x^6 = 280\alpha''\epsilon - 35\beta''\delta + 10\gamma''\gamma - 20\delta''\beta + 24\epsilon''\alpha.$$

Similarly the invariant W and the leading coefficients of X, Y depend on the coefficients of A, G and E^2; and the invariant Z depends on the coefficients of G and E^2.

But these two invariants W and Z have been already calculated; viz. as already mentioned, W is Salmon's invariant D, and Z his invariant E, given each of them in the second edition of his *Higher Algebra* (but not reproduced in the third edition): on account of the great length of these expressions, it has been thought that it was not expedient to give them here.

For the reason appearing above, I have added the expressions for the remaining coefficients of C, E, O.

Leading coefficients of A, B, C, D, E, F, G, H.

$A,\ x^6$	$B,\ x^0$	$C,\ x^6$	$D,\ x^{10}$	$E,\ x^4$	$F,\ x^8$	$G,\ x^{10}$	$H,\ x^{12}$
$a\ \ +1$	$ag+1$	$ae+1$	$ac+1$	$aeg+1$	$ace+1$	$a^2g\ \ +1$	a^2b+1
	$bf-6$	a^2bd-4	a^2b^2-1	$df-3$	d^2-1	$abe-5$	$abc-3$
	c^2+15	c^3+3		e^2+2	a^2b^2e-1	$cd+2$	a^2b^2+2
	d^2-10			$a^2b^2d^2-1$	$bcd+2$	a^2b^2d+8	
				$bcf+3$	c^2-1	bc^2-6	
				$bde-1$			
				c^2e-3			
				cd^2+2			

$I,\ x^0$	$J,\ x^4$	$K,\ x^6$	$L,\ x^{10}$	$M,\ x^6$	$N,\ x^8$	$O,\ x^6$
ace^2+1	$adg+1$	a^2dg+1	a^2bf+1	a^2bf+1	a^2cg-1	a^2cdg
af^2-1	$abef-10$	$aef-1$	$de-1$	$dfg-6$	$deg+1$	$cef-1$
d^2f-1	$cdf+4$	$abcg-3$	$aby-1$	eg^2+8	df^2+3	$df-3$
$dej+2$	ce^2+16	$bdf-2$	$bce-2$	cf^2-3	$ef-3$	b^2-2
e^2-1	d^2e-12	be^2+5	bd^2+4	a^2bg-1	a^2bfg+1	a^2b^2dg
a^2b^2g-1	a^2b^2df+16	$bf+9$	c^2d-1	$bcfg+6$	$bceg+2$	b^2f+1
$bf+1$	b^2e+9	$cde-17$	a^2b^2e+3	$bdeg-34$	bcf^2+3	b^2c^2
$bdg+2$	$bcf-12$	d^2+8	b^2d-6	bcg^2+48	bd^2g-4	$bcdf-14$
$bce-2$	$bcdg-76$	a^2b^2g+2	$bcg+3$	$bef-18$	$bdef-12$	bce^2+11
$bdg-2$	bd^2+48	b^2f-6		c^3g+45	be^2+15	bd^2e+1
bd^2+2	bd^2+48	b^3de+2		c^2df-45	$bef+15$	c^2f+9
c^3e+1	c^3e-32	b^2ce+6		c^2dg+1	c^2dg+9	c^2e-14
c^2d+2	c^2d-32	bcd^2-4		$cdef+78$	$cdf+4$	cd^2+6
c^2d+1				ce^2-36	cde^2-21	a^2b^2f
cd^2e-3				df^2-48	d^3e+8	b^2g+8
d^3+1				c^3d^2+28	a^2b^2cg-3	b^2e^2-9
				a^2b^3ceg	b^3cd+6	b^2cf-6
				b^2d^2g+64	b^2cf+9	$b^2cde+16$
				$b^2def-144$	b^2df+32	b^2d^2-8
				b^2e^2+81	b^2eg-39	bc^3e-3
				b^2cf-96	b^2eg-3	bc^2d^2+2
				b^2c^3+108	b^2fg-3	
				$bcdg+96$	bc^3d-66	
					bc^2f+18	
				$bcde^2-120$	$bcde+76$	
				bd^2e+16	bd^3-32	
				c^4g+36	c^4f+27	
				c^3df-72	c^3de-45	
				c^3e^2-27	c^2d^3+20	
				c^2d^2e+96		
				cd^4-32		

Leading coefficients of P, Q, R.

$P,\ x^2$	$Q,\ x^6$	$R,\ x^6$
$a^2d^2g^2+1$	$a^2dg^2\quad -1$	$+1$
$defy-6$	$efg\ \ +9$	$+9$
df^2+4	$f^2\quad -8$	$+3$
$f^3\quad +4$	a^2bcg^2+3	-24
$g^2y\quad -3$	$bdyg\ -24$	-15
ey^2-3	$be^2g\ -45$	-66
$a\ bcdg^2-6$	$bef\ \ +66$	$-2,\ +3$
$bcefg+18$	$cyf\quad +5,\ +48$	-3
bcf^2-12	$cdeg\ \ +5,\ +48$	-12
bd^2fg+12	$cdf\quad +6,\ -12$	-12
bde^2g-18	$cef\ \ -7$	-51
$bey\ \ +6$	$d^3\quad -3$	-16
$cy\quad +4$	$d^2ef\ -3$	$+36$
c^2e^2g-24	$d^2\quad +4$	-8
c^2fl^2-18	$a^2by^2g^2\ -2$	-2
$c^2ef\ +30$	$b^2g^2\ +4$	$+12$
cd^2eg+54	$b^2deg\ -5$	$+192$
cdf^2-12	$b^2df\ -6$	-48
$cdeg^2-42$	$b^2e^2\ \ +7$	-144
ce^3+12	$bc^2eg\ -5$	-159
$dy\ \ -20$	$bcd^2f\ -6$	$+18$
$d^2ef\ +24$	$bcd^2g\ +7$	-48
$d^2\quad -8$	$bcdef\ -16$	$+24$
$a^2b^2c^2g+4$	$bce^2\ +23$	$+279$
b^2d^2-12	$bdy\ \ +30$	-48
b^2f^3+8	$bd^2y\ -33$	-84
b^2d^2f-3	$c^2dg\ \ -1$	$+42$
b^2d^2+30	$c^2ef\ \ +36$	$+153$
$b^2cef-24$	$c^2dy\ \ -37$	-36
b^2d^2eg-12	$c^2d^2\ \ -53$	-399
$b^2c^2y^2-24$	$cd^3e\ +79$	$+312$
b^2de^2y+60	$d^5\quad -24$	-64
b^2c^4-27	$a^2b^3y\ -2$	
bcy^2d+6	$b^2ceg\ +5$	
$bc^2deg-42$	$b^2cf^2\ +6$	
bc^2df^2+60	$b^2c^2g\ +2$	-224
bc^2ey-30	$b^2def\ +22$	$+144$
bcd^2e^2+24	$b^2e^3\ -27$	$+54$
$bcdy-84$	$b^2c^2dg\ -8$	$+336$
bcd^2e+66	$b^2c^2ef\ -39$	-108
$bdy\ \ +24$	$b^2cdy\ \ -50$	$+384$
bd^2e^2-24	$b^2cde^2\ +107$	-684
$c^3ey\ +12$	$bc^3d^2e\ -22$	$+144$
$cy\ \ -27$	$bc^3f\ +3$	-126
cf^2-8	$bc^3dy\ \ +84$	-648
$c^3def+66$	$bc^2d^3\ \ -21$	$+432$
c^3e^3-8	$bc^2d^2h\ -102$	$+564$
c^3dy-24	$bcd^4\ \ +44$	-288
c^3d^2-39	$c^4f\quad -27$	$+270$
cd^4e+36	$c^4de\ +45$	-450
d^6-8	$c^3d^3\ -20$	$+200$

Remaining Coefficients of C, E, O.

C		E	
x^6y^0	xy	x^2y	x^2y^2
$af+2$ $be-6$ $cd+4$ x^0y^2 $ag+1$ $cc-9$ d^2+8	$adg+1$ $aef-1$ $bcg-1$ $bdf-8$ be^2+9 c^2f+9 $cde-17$ d^3+8	a^2g+1 a^2f+2 $ace-19$ ad^2+8 b^2c-6 $bcd+44$ b^3-30	$aeg-7$ af^2-14 $bdg+28$ $bef+42$ c^2g+14 $cdf-168$ ce^2+105
x^2y^2	x^0y^3	xy^4	x^3y
$bg+1$ $cf-6$ $de+4$ x^3y^0 $eg+1$ $df-4$ e^2+3	$aeg+1$ af^2-1 $bdg-3$ $bef+3$ c^2g+2 $cdf-1$ ce^2-3 d^2e+2	$abg+7$ $aef-14$ $ade-14$ b^2f $bce-21$ bd^2+112 c^2d-70	$afg-7$ $beg+14$ bf^2 $cdg+14$ $cef+21$ d^2f-112 de^2+70
x^4y^0	x^3y^1	x^3y^1	x^4y^1
		$acg+7$ $adf-28$ ae^2-14 $bg+14$ $bcf-42$ $bde+168$ c^2e-105	$adg-1$ b^2g-2 $ceg+19$ cf^2+6 d^2g-8 $def-44$ e^3+30
	x^2y^3	x^2y^4	
	adg $aef-35$ $bcg+35$ bdf be^2+105 c^2f-105	bg^2-1 $efg+5$ $deg-2$ df^2-8 $ef+6$	

Note.—In the tables on this page, a has been treated like the other letters; on the preceding pages, the powers of a have been suppressed except in the first of every series of terms containing a common power of a.

The final result is that we have the values of the invariants B, I, P, W, Z and the leading coefficients of the covariants A, C, D, E, F, G, H, J, K, L, M, N, O, Q, R: also the means of calculating the leading coefficients of the remaining covariants S, T, U, V, X, Y.